Detecting Distributed Social States from Multimodal Signals in Group Conversations

Landry S. Bulls & Mark A. Thornton
Department of Psychological and Brain Sciences, Dartmouth College

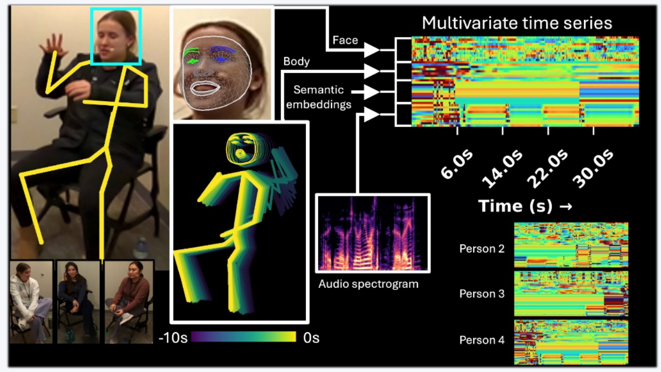
Introduction

Group conversations may transition between awkward silences, excited chatter, and heartfelt storytelling all within the span of a few minutes. This work uses data-driven methods to probe the multimodal indices of such distributed social states recurring across group conversations and understand how they correlate with participants' subjective experiences.

Methods

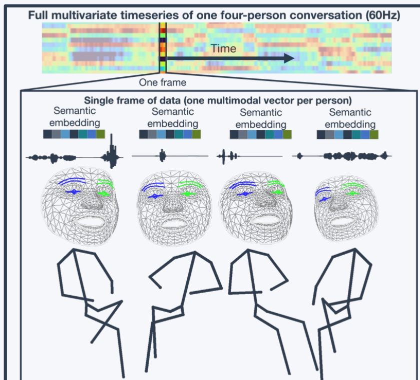
1. Twenty-six unconstrained four-person conversations (N=104) were recorded at 30 minutes each using an array of cameras and microphones
2. Pretrained deep neural network (DNN) models used to extract **facial expressions, 3D body motion, vocal acoustics, and speech semantics** from individuals
3. Collective social states were derived from aggregates of groups' conversational behaviors over **non-overlapping 10-second windows**
4. Uniform manifold approximation (UMAP) and hierarchical density-based clustering (HDBSCAN) were used to discover clusters of recurrent social states
5. Linear mixed-effects models used to predict participants' **post-conversation survey responses** as a function of the **proportion of time spent in each cluster**

Multimodal Feature Extraction



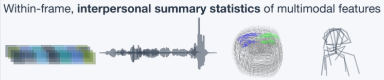
Analysis Pipeline

Full multivariate timeseries of one four-person conversation (60Hz)



Get summary statistics for each feature across people

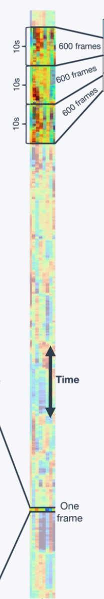
Within-frame, **interpersonal summary statistics** of multimodal features



Semantic1_mean		Voice20_stdev		Face40_skew		Body120_mean
1.202	…	-0.934	…	1.188	…	0.864
-0.094		-0.645		1.004		0.588
0.992		1.99		0.933		0.237

Mean, standard deviation, skew, and kurtosis for each element of each modality's feature vector, describing a summary multimodal state of the group at each frame

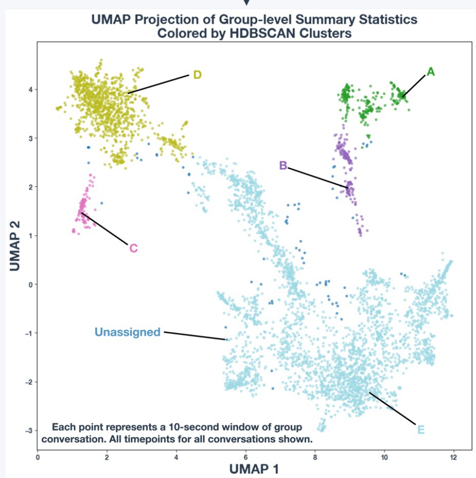
Group-level summary statistic time series

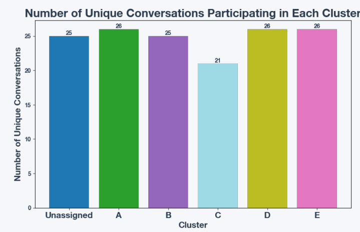


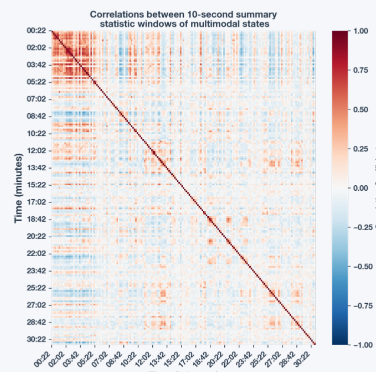
Windowed summary statistics of interpersonal summary statistics

Time window	Face1_mean_mean	Body20_stdev_skew	Voice40_skew_mean	Semantic120_mean_stdev
600s – 610s	2.115	-1.22	1.190	0.064
620s – 630s	-0.185	-0.845	0.630	0.522
630s - 640s	1.46	-1.76	0.618	0.644

10-second windows embedded using 2-D UMAP + Clustering with HDBSCAN

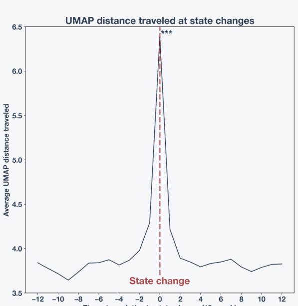


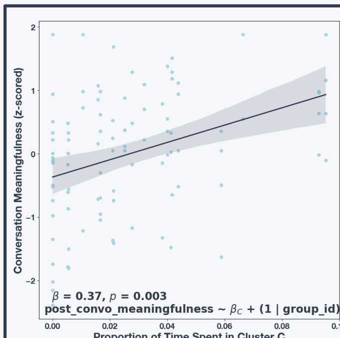


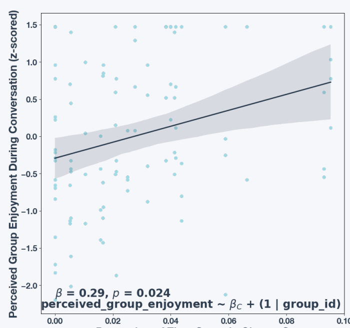


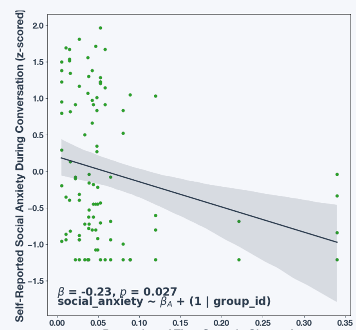
Results

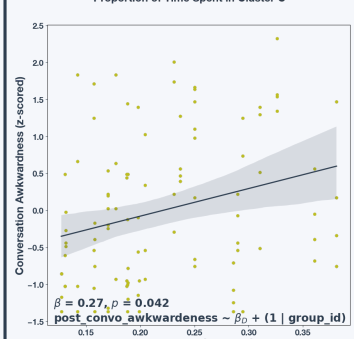
Group conversations exhibit state-like dynamics

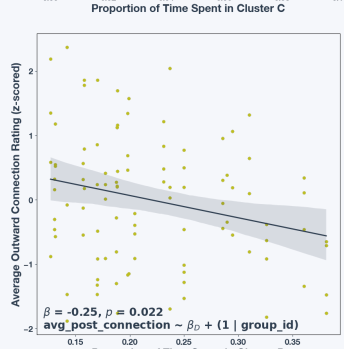












Proportion of time spent in different states predicts post-conversation ratings

Manual inspection of conversational moments from each cluster reveal:

- **Cluster A**: All participants speaking over one another and smiling
- **Cluster C**: One individual recounting personal experiences
- **Cluster D**: Trivial chit-chat; participants pushing conversation forward

Limitations

- This course-grained approach fails to account for meaningful within-person multimodal dependencies, abstracting them to the group level
- Rich information is lost using the windowed aggregation approach
- Difficult to interpret which features are driving different states
- Potentially hierarchical state/event structure is masked using this approach
- Qualitative interpretations of clusters may not be reliable

Conclusions

- Multimodal conversation states recur across different group conversations
- Time spent in different states may predict participants' subjective experience of the conversation and post-conversation attitudes towards one another

Future Directions

- Future work will use methods sensitive to nonlinear dependencies in both intra- and interpersonal multimodal speech
- Post-conversational memory tasks will be used to understand the behavioral indices of collaborative event segmentation

Funding: CompX Grant: Neukom Institute for Computational Science, Dartmouth College